JAN-P-72

30 APRIL 1946

JOINT ARMY-NAVY SPECIFICATION

PRIMER, SYNTHETIC, LAQUER-RESISTING

Army Number	Navy Number
3-183A	52P76

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specification:

U. S. Army	Navy Department
3-183 23 Mar. 1944	

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

A-1. The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN-L-73—Lacquer Enamel, Lustreless.
JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
JAN-P-107—Packaging and Packing for Overseas Shipment—Boxes, Wood, Wire-bound.
JAN-P-124—Packaging and Packing for Overseas Shipment—Cans and Pails, Metal.

U. S. ARMY SPECIFICATIONS

94-40645—Marking, Shipment, Domestic and Export.
100-2—Standard Specification for Marking Shipments by Contractors.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[1]

FEDERAL SPECIFICATIONS

TT-P-141—Paint, Varnish, Lacquer, and Related Materials, General Specification for Sampling and Test Methods.

A-2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.

B. TYPES

B-1. Primer lacquer-resisting covered by this specification shall be of the following types as specified in the contract or order (see par. H-1):

Type I—General purpose.
Type II—Cavity coating.

C. MATERIAL AND WORKMANSHIP

C-1. *Material.*—

[1] Applicable only to Navy purchases.

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C–1a. *General.*—The manufacturer is given latitude in the selection of raw materials and processes of manufacture in order to produce a material of high quality. Rosin, ester gum, and rosin derivatives other than resinate driers will not be permitted.

C–1b. The primer and thinners supplied with or purchased separately shall contain no benzol (benzene) chlorinated or other toxic compounds, or hydrolizable chlorine derivatives.

D. GENERAL REQUIREMENTS

D–1. *Army.*—Unless otherwise specified, primer, synthetic, lacquer-resisting used in finishing equipment for the Army, shall be made in accordance with a formula that has been demonstrated by tests at a Government laboratory, designated by the chief of the supply service concerned, to produce material conforming to the requirements of this specification (see par. H–4).

D–2. *Consistency.*—The consistency of the primer shall be suitable for use by brushing, spraying, or dipping when thinned according to the recommendations of the manufacturer.

D–3. *Keeping properties.*—

D–3a. The primer shall show no excessive settling in a freshly-opened full can, not exceeding 6 months of age, and shall mix readily to a smooth homogeneous state. The primer shall show no curdling, livering, caking, or color separation, and shall be free from lumps and skins.

D–3b. After 30 days standing, a ¾-filled 8-ounce glass jar of the material shall show no livering, curdling, or color separation and shall show no more settling than is easily and readily broken up with a paddle and redispersed. There shall be no hard settlement or caking, and a minimum of skinning. Any skin that may be formed shall be continuous and easily removed.

D–4. *Dilution stability.*—When thinned as specified in paragraph F–4b, the thinner shall not cause the precipitation or separation of any ingredient or affect the quality of the film deposited therefrom.

D–5. *Appearance.*—The primer shall yield a smooth hard satin finish. The film shall not require sanding, and shall be suitable for the application of ammunition lacquer enamel in accordance with the requirements of Joint Army-Navy Specification JAN–L–73, without further treatment.

D–6. *Color.*—No particular color shall be specified other than the natural color obtained by the use of the specified pigments (see par. H–2).

D–7. *Odor.*—The primer shall not have an obnoxious or objectionable odor.

D–8. *Drying time.*—

D–8a. *Baking.*—The primer shall bake hard in 10 minutes at 250° F.

D–8b. *Air-drying.*—

D–8b(1). *Brushing.*—A brush coat of the primer shall set to touch in 20 minutes and dry hard in 30 minutes.

D–8b(2). *Spraying.*—A spray coat of the primer shall set to touch in 4 minutes and dry hard in 8 minutes.

D–8b(3). *Dipping.*—A dip coat of the primer shall set to touch in 6 minutes and dry hard in 10 minutes.

D–9. *System adhesion.*—The system applied and tested as specified in paragraph F–4d shall adhere tightly. After bending and removing the loosened primer, the width of the line of bare steel shall be fairly uniform and average no wider than ⅛-inch. It shall not be possible to expose large spots of bare steel by rubbing with the ball of the thumb across the scratch.

D–10. *Knife test requirement.*—The primer shall adhere tightly to the panel when subjected to the knife test specified in paragraph F–4c. It

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shall be difficult to furrow off the primer with a sharp knife. Flakes shall not fly from the surface, and the scratch shall show beveled edges, indicating excellent adhesion.

D–11. *Flexibility.*—The primer shall show no cracking or flaking at the bend, when tested as described in paragraph F–4f.

D–12. *Immersion resistance.*—

D–12a. *Cold water resistance.*—No wrinkling or blistering shall be visable to the eye immediately upon removal of the panel from the water, and shall be no more than slightly affected when observed 2 hours after removal from the water. After 24 hours air-drying, the portion of the panel which was immersed shall be indistinguishable from the portion which was not.

D–12b. *Hydrocarbon resistance.*—No wrinkling or blistering shall be visible to the eye immediately upon removal of the panel from the hydrocarbon mixture, and shall be no more than slightly affected when observed 2 hours after removal from the hydrocarbon mixture. After 24 hours air-drying, the portion of the panel which was immersed shall be indistinguishable from the portion which was not.

D–13. *Salt spray resistance.*—A film of the primer of 0.75 mil maximum dry thickness, shall pass the test specified in paragraph F–4g. Panels shall show no undercutting of the coating or any evidence of rusting at points other than within ⅛-inch of the scribe marks or edges of panel. Panels shall show no blistering or other defects of the primer film.

D–14. *Lacquer resistance.*—The primer shall withstand recoating with lacquer enamel without lifting, bleeding, or being dissolved, when tested as specified in paragraph F–4h.

E. DETAIL REQUIREMENTS

E–1. *Composition.*—

E–1a. *Total solids content.*—Total solids content shall be a minimum of 60 percent, by weight.

E–1a(1). *Vehicle solids content.*—The solids content of the isolated vehicle shall be a minimum of 28.0 percent, by weight.

E–1a(2). *Pigment volume.*—Pigment solids content shall be a maximum of 47.0 percent of the total solids content, by volume.

E–1b. *Total vehicle content.*—

E–1b(1). *Volatile vehicle content.*—The volatile content of the primer shall consist of any suitable solvent or solvents.

E–1b(2). *Non-volatile vehicle content.*—The resin used in this primer shall be a blend of oil modified alkyd resin, modifying agent and varnish resin or oleoresinous varnish in the following proportions by weight (see par. H–3):

	Percent
Type I [1] Oil modified alkyd resin (30 percent phthalic anhydride, min.)	70
Varnish resin (max.)	20
Modifying agent (max)	10
Type II Oil modified alkyd resin (35.0 percent phthalic anhydride, min.)	70
Phenolaldehyde resin, dispersion type	30

E–1b(3). The vehicle solids content shall conform to the following percentage by weight:

	Type I	*Type II*
Phthalic anhydride (min.)	21.0	24.5
Oil acids (min.)	31.5	25.8

E–1b(4). The resin shall show a negative rosin test.

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[1] Shorter oil length may be used, the balance of the oil being added with the varnish resin as oleoresinous varnish.

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E-1c. *Pigments.*—The pigmentation shall be as follows:

	Percent
[1] Zinc chromate pigment	7.0—20.0
Iron oxide red (Fe_2O_3 by analysis min.)	35.0
Zinc oxide (max.)	20.0
Extenders (including that from iron oxide max.)	58.0

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F-1. *Size of lot.*—Unless otherwise specified, a lot shall consist of primer from one batch submitted at one time for inspection.

F-2. *Sampling.*—A composite sample of approximtely 1 quart shall be taken in accordance with the requirements of Federal Specification TT-P-141. The sample shall be labeled to show the name of the material, manufacturer, plant, contract or order number, lot number, number of gallons in the lot, and the proportions of enamel and thinner recommended by the manufacturer as follows:

Thin—volume of primer with —volume of thinner for brush application.

Thin—volume of primer with—volume of thinner for spray application.

Thin—volume of primer with—volume of thinner for dip application.

All acceptance tests shall be made on the sample representative of the lot.

F-3. *Inspection.*—Inspection shall be made in accordance with the requirements of Federal Specification TT-P-141.

F-4. *Tests.*—All tests shall be made in accordance with the requirements of Federal Specification TT-P-141, except as otherwise stated in the following paragraphs:

F-4a. *Preparation of test panels.*—Ten solvent-cleaned steel panels shall be sprayed, except for panels required for the brushing and dipping requirements specified in paragraphs D-8b(1) and D-8b(3) to a maximum dry film thickness of 0.75 mil. One of these panels shall be used immediately for the drying time test. Two panels shall be air-dried for 24 hours, and the remaining 7 shall be baked at 150° ±5° F. for 24 hours and used where aged panels are required.

F-4b. *Dilution stability.*—Thin approximately 1 pint of the enamel by mixing thoroughly with 25 percent more than the recommended amount of thinner. Note if there is any precipitation or separation of ingredients. Pour out the thinned enamel on a solvent-cleaned panel and air-dry.

F-4c. *Knife test.*—The primer shall be cut from an aged panel and tested in accordance with the requirements of Federal Specification TT-P-141.

F-4d. *System adhesion.*—Spray an aged panel with olive drab lacquer enamel conforming to the requirements of Joint Army-Navy Specification JAN-L-73 and allow to air-dry 2 hours. Cut a fine line across the panel with a razor blade, penetrating to the steel. Bend the panel rapidly through 180 degrees over a ⅜-inch mandrel, coated side out, with the scratch directly over and parallel to the mandrel. Rub across the scratch vigorously with the ball of the thumb removing all loosened material (do not use finger nails or knife). Measure the width of the bared metal.

F-4e. *Immersion tests.*—Aged panels shall be immersed for 18 hours in the immersion fluid at 25° C. (77° F.).

F-4e(1). *Water resistance.*—Distilled water shall be used.

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[1] Two types of zinc chromate pigments are acceptable in compliance with the requirements of this specification: Zinc yellow, 41.0 percent minimum CrO_3; and zinc tetroxy chromate, 17.0 to 18.0 percent CrO_3. (The manufacturer shall state which type is used.)

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F–4e(2). *Hydrocarbon resistance.*—A petroleum fraction or mixture having the following characteristics shall be used:

(a) Mixed aniline point....50° to 60° C.
(b) Specific gravity.....0.725 to 0.750 at 25/25° C.
(c) Boiling range.....The fraction shall boil completely within the range of 32° to 204° C. The first drop shall distill not higher than 46° C.

F–4f. *Flexibility.*—Bend an aged panel rapidly through 180 degrees over a ⅜-inch mandrel, coated side out, and examine with a 5-power glass.

F–4g. *Salt spray resistance.*—Two aged panels shall be used in a 48-hour test.

F–4h. *Lacquer resistance.*—Spray 5 clean panels with a coat of the primer and air-dry one for 10 minutes: one for 6 hours; one for 24 hours; and one for 48 hours. At the end of these periods of time, spray with a topcoat of white test lacquer. Observe for lifting or dissolving on all panels as evidenced by blistering, wrinkling, discoloration of the white lacquer, or other film irregularities. The nitrocellulose white test lacquer shall be of the following composition:

	Percent
Nitrocellulose	5.0
Ester gum	4.0
Blown castor oil	2.0
Titanium dioxide	6.0
Ethyl acetate	15.0
Ethyl alcohol	14.0
Butyl acetate	12.0
Toluol	42.0

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging (for Army Air Forces and for Navy purchases only).*—

G–1a. *For domestic shipment.*—Unless otherwise specified, primer shall be furnished in 1-gallon multiple-friction top cans or in 5-gallon lug-cover steel pails conforming to the requirements of Joint Army-Navy Specification JAN–P–124, except that exterior protective coatings will not be required on 1-gallon multiple-friction top cans (see par. H–1).

G–1b. *For overseas shipment.*—Unless otherwise specified, primer shall be furnished in 1-gallon multiple-friction top cans or in 5-gallon lug-cover steel pails conforming to the requirements of Joint Army-Navy Specification JAN–P–124. After filling, the plugs shall be spot-soldered to the friction rings at three points spaced equidistant from each other around the periphery of the plug.

G–2. *Packing.*—

G–2a. *Army.*—Packing for Army shall be in accordance with the requirements of Federal Specification TT–P–141.

G–2b. *Navy*—

G–2b(1). *For domestic shipment.*—Unless otherwise specified, primer packaged as described in paragraph G–1a shall be packed with four 1-gallon cans in a nailed wood or wirebound box, conforming to the requirements of Joint Army-Navy Specification JAN–P–106 or JAN–P–107, respectively, or in a style RSC corrugated or solid fiberboard box, having a minimum Mullen test of 275 pounds. No strapping will be required. Unless otherwise specified, 5-gallon lug-cover steel pails will require no further packing.

G–2b(2). *For overseas shipment.*—Unless otherwise specified, primer packaged as described in paragraph G–1b, shall be packed with four 1-gallon cans in a nailed wood or wire-bound box, conforming to the

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requirements of Joint Army-Navy Specification JAN–P–106 or JAN–P–107, respectively. Boxes shall be strapped in accordance with the requirements of the appendix of the applicable box specification. Unless otherwise specified, 5-gallon lug-cover steel pails will require no overpacking.

G–3. *Marking.*—In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2 for Army Service Forces and in accordance with U. S. Army Specification 94–40645 for Army Air Forces; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. Requests, requisitions, schedules, and contracts or orders should contain the following features:

(a) Title of the specification, the number and date.
(b) Type of primer required (see par. B–1).
(c) Whether packaged and packed for domestic or overseas shipment (see sec. G).
(d) Size of container required (see par. G–2a.).

H–2. It is important to note that the color will vary according to the type and amounts of color pigments used.

H–3. *Use.*—Type I and type II synthetic primer are intended for use as the first coat on bare metal surfaces, with an enamel or a lacquer enamel as topcoat.

H–4. Information concerning tests to demonstrate the acceptability of paints for use on Army Ordnance materiel may be obtained from the Chief of Ordnance, Army Service Forces, Washington 25, D. C. or the Paint and Chemical Laboratory Research and Development Center, Aberdeen Proving Ground, Md.

H–5. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, U. S. Army, and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–6. Copies of Joint Army-Navy and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–7. Copies of this specification (required for Army purchases) may be obtained from the office, Chief of Ordnance, Army Service Forces, Washington 25, D. C.

H–8. Copies of this specification (required for Navy purchases) may be obtained as indicated in paragraph H–6.

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